

Brazilian Synchrotron Light Laboratory

Gabriel Rezende

Overview of ID tracking methods

Campinas - SP
September 2026

Contents

1 Introduction

This document presents a detailed overview of insertion devices tracking methods. The first section is an introduction to the Ellaume formalism, derivations and predicted effects on the beam. The second section presents the tracking method based on Runge-Kutta integration and compares results with those obtained from the Ellaume formalism. Finally, last section presents an symplectic integration method proposed by Wu, Forest and Robin and also compares the results with the two first methods. The main goal of this document is to provide a complete overview of the methods used to track insertion devices in the SIRIUS storage ring and to compare their results.

2 Ellaume formalism

2.1 Equations of motion

Using SIRIUS coordinate system, the equations of motion for a particle in a magnetic field can be expressed as:

$$\frac{d\vec{p}}{dt} = q\vec{v} \times \vec{B} \quad (1)$$

$\vec{p}$ is the momentum of the electron, $q = -e$ is the electron's charge, $\vec{v}$ is its velocity, and $\vec{B}$ is the magnetic field along the electron's path. The velocity can be written as:

$$\vec{v} = \frac{dx}{dt}\hat{x} + \frac{dy}{dt}\hat{y} + \frac{ds}{dt}\hat{s} = \left(\frac{ds}{dt}\right) \left(\frac{dx}{ds}\hat{x} + \frac{dy}{ds}\hat{y} + \hat{s}\right) \quad (2)$$

defining

$$x' := \frac{dx}{ds}, \quad y' := \frac{dy}{ds} \quad (3)$$

and assuming the electron is ultra-relativistic, we have:

$$c^2 = \vec{v}^2 = \left(\frac{ds}{dt}\right)^2 (x'^2 + y'^2 + 1) \quad (4)$$

then, it is found the following expression for ds/dt :

$$\frac{ds}{dt} = \frac{c}{\sqrt{x'^2 + y'^2 + 1}} \quad (5)$$

which gives the expression for the velocity:

$$\vec{v} = \frac{c}{\sqrt{x'^2 + y'^2 + 1}} (x'\hat{x} + y'\hat{y} + \hat{s}) \quad (6)$$

Substituting this expression into the Lorentz force equation, we have:

$$\frac{d\vec{p}}{dt} = \frac{-ec}{\sqrt{x'^2 + y'^2 + 1}} [(y'B_s - B_y)\hat{x} - (x'B_s - B_x)\hat{y} + (x'By - y'Bx)\hat{s}] \quad (7)$$

Keeping the result of equation (??), now let's express the momentum of a relativistic electron:

$$\vec{p} = \gamma m_e \vec{v} = \frac{\gamma m_e c}{\sqrt{x'^2 + y'^2 + 1}} (x'\hat{x} + y'\hat{y} + \hat{s}) \quad (8)$$

and it's time derivative:

$$\frac{d\vec{p}}{dt} = \frac{ds}{dt} \frac{d\vec{p}}{ds} = \frac{\gamma m_e c^2}{\sqrt{x'^2 + y'^2 + 1}} \left[-(x'^2 + y'^2 + 1)^{-3/2} (x'x'' + y'y'') (x'\hat{x} + y'\hat{y} + \hat{s}) + \frac{x''\hat{x} + y''\hat{y}}{\sqrt{x'^2 + y'^2 + 1}} \right] \quad (9)$$



Figure 1: Field roll-off for different block widths

2.2 Comparisons between roll-offs

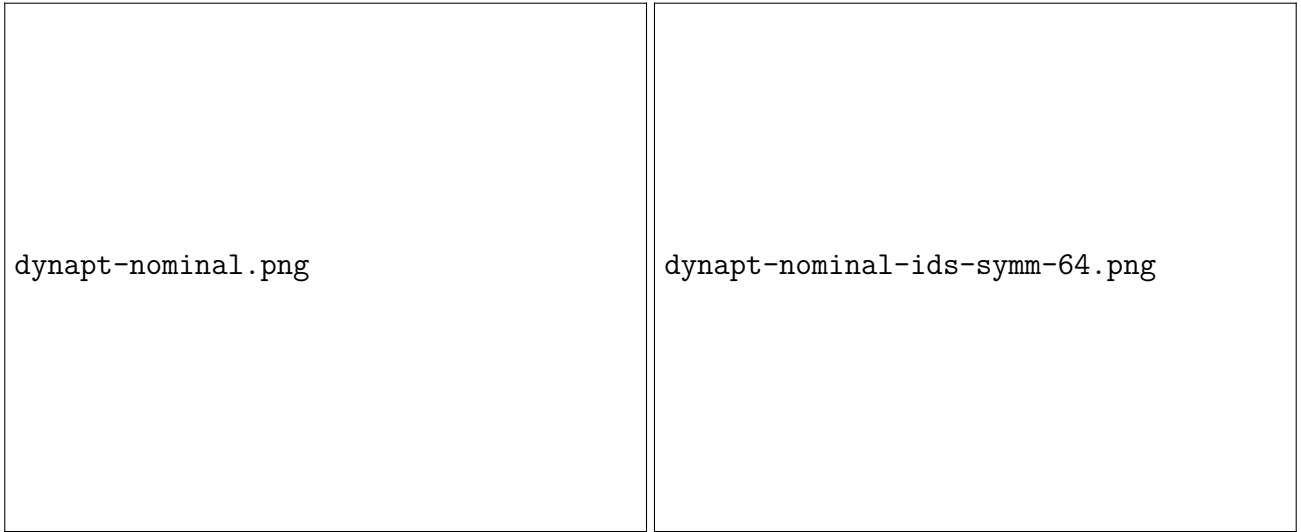


Figure 2: The left image shows the nominal dynamic aperture, while the right image shows the dynamic aperture after the insertion of an IVU18 with 64 mm block widths.

3 Conclusion

With the results shown here, it is possible to conclude that the biggest effect on the dynamic aperture happens after the insertion of the IVU18 and does not change considerably when the widths of the blocks are reduced to the values considered here, therefore, the reduction of the roll-off specification for 0.5% will not affect the dynamic aperture.